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MACHINE LEARNING (ML)

4 Types
- ↳ Supervised learning
- ↳ Unsupervised learning
- ↳ Semi supervised learning
- ↳ Reinforcement learning

① ONLINE LEARNING :-

→ Datasets that arrives continuous and it can be trained without putting down the server.

Working :- ① Receives new data.
② Update Model
③ Make prediction
④ Repeat.

Apr Recommendation Systems

② BATCH LEARNING :-

A ML is trained on the entire dataset at once.

Working :- ① Collect data.
② Train model on all data.
③ Deploy Model.
④ Retrain when new data arrives.

Apr Neural Networks

Instance-Based Learning	Model Based Learning
→ Stores exp.	→ Learns formula
→ Lazy learning	→ Eager learning
→ Fast training	→ Slow training
→ Slow prediction	→ Fast prediction
→ High Memory	→ Low memory

TENSORS

→ A tensors is multidimensional array used to represent data in ML and DL.

Tensors = Generalization of scalars, vectors, and Matrices.

↳ 0D → 5
↳ 1D → [1,2,3]
↳ 2D → [[1,2,3],[4,5,6]]
↳ 3D → RGB Image
↳ (3,224,224).
↳ 4D → (32, 224, 224, 3)

How to frame a ML problem:

Business problem to ML Problem.

↓

Types of Problem

↓

Current Soln.

↓

Getting Data.

↓

Metrics to Measure

↓

Online vs Batch

↓

Check Assumption.

Ex:
Predict the customer about to Churn from Netflix

Data Gathering:

- CSV/ excel
- JSON/ SQL
- fetching from API
- Web Scraping.

→ CSV/ Excel: used when data is already available as files

→ **JSON:**

Used for semi-structured data, mostly from APIs.

→ **SQL:**

Used when data is stored in databases

→ **APIs:**

Used to fetch real time or live data.

→ **Web Scraping:**

Used only when no APIs is available.

Understand your data:

- Asking basic questions
- EDA (univariate)
- EDA (Multivariate)
- Pandas profiling.

① **Asking basic Questions:**

- How big is your data? → df.shape.
- How does the data look like? → df.head().

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- What is the data type of cols? → df.info().
- Are there any missing values? → df.isnull().sum().

- How does the data look mathematically? → df.describe().
- Are there duplicate values? → df.duplicate.sum()
- How is the correlation b/w columns? → df.corr().

② EDA(univariate):

→ One variable at a time. Basically how does this single column look like.

→ Check: distribution, central tendency, Outlier, Missing Values.
→ Common Visuals: Histogram, Box plot.

③ EDA (Bivariate):

→ Two variables together
→ How one column change with another?
→ Check: Correlation, Trends
→ Common Visuals: Scatter plot.

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Multivariate

- More than 2 variable.
- How do multiple columns interact together.
- Checks combined effects, patterns.
- Common visuals - Heatmaps, pair plot.

4. Pandas profiling

- An automated EDA tool that summarizes the dataset for you.
- Helps you quickly understand structure & quality of data
- Useful for initial exploration, not final conclusions.

WHAT IS FEATURE ENGINEERING?

- Feature engineering is the process of using domain knowledge to extract features from raw data.
- These features can be used to improve the performance of machine learning algorithms.

FEATURE ENGINEERING.

- Feature Transform-ation
  - Missing value imputation.
  - Handling Categorial features.
  - Outlier Detection
  - Feature Scaling.
- Feature Construction
- Feature Selection
- Feature extraction

Feature transformation

① **Missing Value imputation :-**

Either remove or fill missing values. Replace by mean, median or most frequent category.

② **Handling Categorical features :-** eg:- One-hot encoding.

③ **Outlier Detection :-** Outliers removed are much better fit.

④ **Feature Scaling :-** eg:- Age in range of 100s
Salary in range of 40000s

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→ So for KNN eg → $\sqrt{(X_2-X_1)^2+(Y_2-Y_1)^2}$

salary form
dominations.

→ Thus we bring all values to common scale
→ Generally b/t -1 to 1

Feature Construction:-

→ Create new column manually using domain knowledge.
→ eg:- New column by adding 2 column.

Feature Selection:-

→ Selecting the features useful for the model.
→ Speed of the model increases.

Feature extraction:-

→ eg:- PCA (Principal Component Analysis), LDA
→ This is done by system, not manually.